PromoBase

Product Requirements Document

Chrome extension first release

A personal library for promo codes, with a timely reminder when a saved merchant site is visited.

Document field	Value
Version	0.3 - MIT license decision recorded
Status	Ready for product and implementation planning
Date	8 October 2026
Product owner	Andrii Marusiak
Primary audience	Product owner, design, engineering, quality assurance
First release	Chrome desktop extension, guest-only local storage

This document defines the first buildable version. It records product behavior, user-visible rules, technical choices, acceptance criteria, and future ideas so that product, design, and engineering share the same baseline.

Contents

1. Product summary
2. Problem and product opportunity
3. Goals and non-goals
4. Users and primary workflow
5. Scope and release boundaries
6. Product behavior and requirements
7. User interface and interaction rules
8. Data model and business rules
9. Technical design and stack
10. Privacy, security, and permission model
11. Market comparison and fit hypothesis
12. Success measures and validation plan
13. Risks and mitigations
14. Delivery approach
15. Future improvements
16. Decisions and assumptions
17. Glossary
18. Additional Notes
19. References

Note: Headings use Word's built-in heading styles so the document outline remains navigable in Microsoft Word and Google Docs.

1. Product summary

PromoBase is a Chrome extension that lets a person save promo codes they encounter online, organize those codes in a personal list, and receive a reminder when they visit the associated merchant website. Users enter codes manually. Creator-sponsored codes are the motivating use case, while the product remains useful for any promo code a person wants to remember.

The first release is deliberately small: no account, no server, and no sharing. Records live in the user's Chrome extension storage. When the user visits a saved merchant domain, PromoBase can open its toolbar popup and show matching codes. PromoBase requests permission only for sites where the user has left Notify enabled.

The product is technically feasible for a solo developer with QA experience. The key implementation risks are site-permission friction, popup interruption, and correct merchant-domain matching. The recommended stack is TypeScript with Manifest V3 for the extension, React for the interface, and Chrome extension storage. Python with FastAPI and PostgreSQL is reserved for a later account and synchronization phase.

At a glance	Initial decision
Primary job	Remember a code and surface it at the merchant site when useful.
Target platform	Google Chrome desktop, Manifest V3.
Account requirement	None for the first release.
Storage	chrome.storage.local on the current Chrome profile.
Free record limit	100 saved records.
Reminder	PromoBase toolbar popup, opened on a matching site visit.
Market-fit hypothesis	4 out of 10 today; a focused niche with demand to validate.

2. Problem and product opportunity

2.1 User problem

People hear or see promo codes in creator videos, streams, podcasts, newsletters, and other online content. They may not need the advertised product at that moment. A note in a separate app can be forgotten, and the user has to remember which merchant the code belongs to when they eventually shop.

2.2 Product opportunity

PromoBase provides a lightweight personal memory layer. The user records a code and its merchant once. Later, the extension recognizes the merchant site and presents the saved code without requiring the user to search through notes or remember the code. The user remains in control of which merchant sites PromoBase can access.

2.3 Product positioning

PromoBase is a personal promo-code organizer with context-aware reminders. It is not a coupon discovery engine, affiliate shopping portal, price-comparison tool, or automatic checkout optimizer. Its initial value comes from remembering codes the user already found.

3. Goals and non-goals

3.1 First-release goals

- Let a guest add, view, edit, and delete their own promo-code records without registration.
- Make saving a code quick enough to use while watching a video or browsing another page.
- Publish the source code in a public GitHub repository with a clearly selected open-source license and safe contribution guidance.
- Match a saved merchant registrable domain across its root hostname and subdomains.
- Open a PromoBase toolbar popup when the user visits a matching site and has allowed the required site access.
- Keep expired codes discoverable, clearly marked, and lower in the list rather than deleting them.
- Explain local data loss plainly and keep the guest limit visible.

3.2 Non-goals for the first release

- Creating an account, cloud backup, cross-device sync, or recovery after removing the extension.
- Sharing codes with other users or receiving shared codes.
- Finding, scraping, validating, testing, or auto-applying codes from the web.
- Reading or analyzing video content, creator descriptions, browsing history, or page text beyond the minimum URL needed for a configured reminder.
- Subscription billing, embedded advertisements, donations, affiliate tracking, or in-extension feedback submission.
- Native mobile apps, Firefox, Edge, Safari, or Chrome mobile support.

4. Users and primary workflow

4.1 Primary user

A Chrome user who shops online and wants to save discount codes found from creators or elsewhere, without opening an account. The initial product owner is also the first likely power user and quality reviewer.

4.2 Primary workflow

1. The user hears a code such as QWERTY in a video and opens PromoBase from the Chrome toolbar.
2. The user enters the code and, when available, the merchant URL or domain, any dates or note. Notify is checked by default only when a usable merchant URL is present.
3. PromoBase validates the fields, derives the merchant registrable domain, saves the record locally, and requests site access if required for reminders.
4. Later, the user visits the merchant website. PromoBase finds eligible saved records for that merchant and opens its toolbar popup.
5. The popup displays each matching code, validity context, and an action to copy the code. The user can dismiss the popup and continue browsing.

4.3 Example record

Field	Example value	Behavior
Promo Code	QWERTY	Required. Trimmed text value.
Resource or Product URL	https://www.dropbox.com/	Optional. Used to derive the merchant domain.
Start Date	Not set	Optional. If set, reminders begin on that local calendar date.

Field	Example value	Behavior
Expiry Date	31.12.2026	Optional. Valid through the end of that date in the current timezone.
Note	2 TB plan discount	Optional. Maximum 140 characters.
Notify	Enabled when a usable merchant URL is supplied	Checked by default only when a usable URL is present. Otherwise off and disabled.

5. Scope and release boundaries

5.1 Included in the first release

- Guest onboarding with an always-visible data-retention warning.
- A local list with a 100-record maximum, record count, search, and active/expired ordering.
- Create, edit, and delete operations for code records.
- Manual entry and validation of all record fields.
- Merchant domain normalization and reminder matching across subdomains.
- A permission-aware toolbar popup reminder with code copying and expiry warnings.
- Current-timezone date display and local calendar-day date semantics.
- Accessible labels, keyboard-operable controls, and clear empty/error states.
- Public GitHub repository with a selected license, setup instructions, and no secrets or real user data.

5.2 Explicitly deferred

Accounts, cloud storage, sharing, monetization, and donations remain future options. They are not prerequisites for learning whether the save-and-remind behavior is useful.

6. Product behavior and requirements

The identifiers below are stable references for implementation tasks and later acceptance testing. All requirements in this section define the initial release scope unless explicitly deferred elsewhere in this document.

PB-001 Guest access

A user can start using PromoBase without creating an account.

Acceptance criteria

- First launch opens a concise welcome view with a clear Add a code action.
- No email address, password, or sign-in step is required to create or manage local records.
- The interface identifies the current storage mode as guest and local.

PB-002 Data-retention warning

A persistent warning explains where guest records are stored and what can remove them.

Acceptance criteria

- The warning is visible in the main extension interface and remains available in settings or an About view.
- Use this plain-language wording or an equivalent: Your promo codes are saved in this Chrome profile. Clearing browsing history will not remove them. If you remove PromoBase or reset or lose this Chrome profile, your saved codes may be lost. This version does not include cloud backup.

- Do not claim that clearing Chrome browsing data clears extension records. Chrome documents extension storage as persisting through browsing-data clearing and being cleared when the extension is removed [R1, R2].

PB-003 Record capacity

Guest users can store up to 100 records.

Acceptance criteria

- The list shows the number of saved records and the 100-record limit.
- The 100th record can be saved. The next save is blocked with a clear explanation and next step.
- Editing an existing record remains possible at the limit.
- Deleting a record releases one slot.

PB-004 Create a record

The user can enter and save a promo-code record with the fields listed in Section 8.

Acceptance criteria

- Promo Code is required and cannot be blank after trimming leading and trailing whitespace.
- Resource or Product URL, Start Date, Expiry Date, and Note are optional.
- When a valid Resource or Product URL is present, Notify is checked by default for a new record.
- When the Resource or Product URL is empty, Notify is off and disabled. The record can still be saved, but cannot trigger a site-visit reminder.
- If a user removes the URL while editing a record, disable Notify and turn it off; remove reminder access if no other enabled record needs it.
- Note is limited to 140 characters and the interface communicates the remaining or used character count.
- Invalid URLs and invalid date combinations receive actionable inline messages.

PB-005 Merchant domain normalization

PromoBase converts an entered URL or hostname to a canonical registrable domain for matching.

Acceptance criteria

- Accept a full URL such as https://www.dropbox.com/plans, or a domain such as dropbox.com.
- Ignore scheme, path, query, fragment, port, and a leading www when deriving the matching key.
- www.dropbox.com, dropbox.com, and shop.dropbox.com resolve to the same saved merchant key dropbox.com.
- Match on DNS label boundaries. dropbox.com.example.org and notdropbox.com do not match dropbox.com.
- Use Public Suffix List-aware registrable-domain parsing so suffixes such as co.uk are handled correctly [R13].

PB-006 Site permission

PromoBase requests access only for saved merchant domains where Notify is enabled.

Acceptance criteria

- The permission explanation states that PromoBase checks visits to the saved site only to show matching saved codes.
- The permission request is initiated from a clear user action, such as saving a record with Notify checked, consistent with Chrome's runtime permission model [R4].
- If permission is declined, the record is still saved and can be opened from the extension, while the record indicates that automatic reminders need site access.
- The user can later enable access and reminders from the record or settings.

- Turning Notify off removes PromoBase's need to monitor that merchant for reminders and should unregister its content script or access where Chrome permits.

PB-007 Visit reminder

When the user reaches a saved merchant site, PromoBase opens its toolbar popup with matching codes.

Acceptance criteria

- Reminder is triggered on a top-level visit to a matching HTTP or HTTPS registrable domain for which Notify and site access are both enabled.
- A single popup groups all matching eligible records rather than opening one popup per code.
- The popup shows the promo code, merchant, note when present, start/expiry context, and a Copy code action.
- When multiple records match, show all of them in one popup and order non-expired records before expired records.
- Show at most one automatic popup per registrable domain visit per tab. Do not reopen on reloads or same-domain route changes; allow a new reminder after the user leaves that registrable domain and later returns.
- If a popup cannot be opened or the site permission is unavailable, do not lose the record. Keep it accessible from the extension action and use an unobtrusive badge where practical.
- Chrome exposes action.openPopup from Chrome 127 onward; this is the implementation basis for a toolbar popup reminder [R5, R6].

PB-008 Validity and expiry

Expiry affects display and warning text but does not silently remove records or suppress visit reminders.

Acceptance criteria

- A code with no start or expiry date is treated as valid indefinitely for list ordering and reminders.
- A future Start Date prevents reminders until the beginning of that date in the current local timezone. Before then, show the record as Upcoming in the list.
- An Expiry Date is inclusive through 23:59:59.999 on that local calendar day. Starting the next day, mark the code Expired.
- Expired records remain visible, appear gray or otherwise visually subdued, and sort after non-expired records.
- Visiting the merchant for an expired code still opens the reminder and labels it clearly, for example: This code may have expired. Its saved expiry date was 31.12.2026.
- If Start Date is later than Expiry Date, block save and explain the conflict.

PB-009 List and find

The user can review saved records and find them by code or merchant.

Acceptance criteria

- Show merchant, code, date status, note preview, and Notify state in each row or card.
- Search is case-insensitive and matches code, normalized merchant domain, and note.
- Sort active and undated records first, upcoming records next, and expired records last. Within a status group, newest records appear first.
- An empty state explains how to add the first code. A no-results state offers a way to clear search.

PB-010 Edit a record

The user can change any saved field without changing the record identity.

Acceptance criteria

- Opening edit shows the current saved values.

- The same validation and date rules as “create” apply.
- Changes are saved to local storage and reflected immediately in the list and reminder registration.

PB-011 Delete a record

The user can delete a record they no longer need.

Acceptance criteria

- Delete requires a confirmation step or a reversible undo affordance.
- After deletion, the record no longer appears in list results or future reminders.
- Its site registration or permission is removed only if no remaining Notify-enabled record needs that merchant domain.

PB-012 Copy code

The user can copy a displayed code with one action.

Acceptance criteria

- Copy uses the browser clipboard when available and displays a short success message.
- If copying fails, the code remains selectable and a useful fallback message is shown.

PB-013 Local persistence

Records are saved using Chrome extension storage and survive ordinary browser restarts and browsing-data clearing.

Acceptance criteria

- Use chrome.storage.local rather than localStorage in a content script.
- Records remain after closing and reopening Chrome and after clearing ordinary browsing data.
- Records are expected to be removed when PromoBase is uninstalled or the Chrome profile is reset or lost [R1, R2].
- The initial release does not promise cross-device sync, backup, or recovery.

PB-014 Save and error recovery

PromoBase explains failures clearly and protects the user's unsaved work when an operation cannot complete.

Acceptance criteria

- Do not show a success state unless the storage operation has completed successfully.
- If a create or edit save fails, keep the entered values on screen and show a plain-language message with a retry action where retry is safe.
- If Chrome storage is unavailable or rejects a write, do not clear the form or imply that the record was saved. Keep existing saved records visible when possible.
- If validation fails, identify the field and explain what the user can correct. Do not treat a missing optional URL as a save error.
- If site permission is denied or cannot be requested, save the record if local storage succeeds, preserve the user's Notify selection, and explain that reminders are unavailable until access is enabled.
- If a matching-site popup cannot open, preserve the record and provide access from the toolbar action or library; do not repeatedly retry during the same visit.
- Do not transmit error details, promo codes, notes, or visited URLs to a server in the first release.

PB-015 Public source repository

PromoBase source code is maintained in a public GitHub repository under an explicit open-source license.

Acceptance criteria

- Before public release, add an MIT license file and identify MIT in the README.

- README explains the project purpose, supported Chrome version, local data behavior, installation for development, and how to report issues.
- Do not commit credentials, signing keys, private test data, or real user promo-code records. Use synthetic examples in documentation and tests.
- Review third-party dependency licenses before distribution and keep the Chrome Web Store privacy policy and data-use disclosures accurate [R3, R14, R15].

7. User interface and interaction rules

7.1 Primary surfaces

Surface	Purpose	Key content
Toolbar popup	Quick add and entry point to the library.	Add code, search/list summary, record limit, local-data warning.
Full library view	Review and maintain saved records.	Search, status grouping, edit, delete, and reminder permissions.
Automatic reminder popup	Surface codes relevant to the current merchant.	Matching codes, copy actions, date status, explicit expired warning.
Permission explanation	Explain site access at the point Notify is enabled.	Merchant domain, why access is needed, and how to proceed if declined.

7.2 Interaction principles

- Keep the “save” flow compact. Required fields should fit in the initial view, with optional fields visible without making the task feel heavy.
- Do not interrupt a site visit with a generic permission prompt. Explain the matching behavior before the user grants access.
- Show expired status in both the library and the reminder popup. Never present an expired code as confirmed valid.
- Use clear labels such as Upcoming, Active, and Expired rather than relying on color alone.
- Support keyboard focus, visible focus indicators, readable contrast, and labels that screen readers can announce.

7.3 Repeat reminder policy

The initial behavior is one automatic reminder per registrable-domain visit per tab. Refreshes and same-domain navigation should not reopen it. Leaving the merchant domain and returning starts a new visit. This balances visibility with avoiding repeated interruptions.

8. Data model and business rules

8.1 Record fields

Field	Type	Required	Rule
id	UUID string	Yes	Stable record identifier generated on creation.
promoCode	String	Yes	Trimmed, non-empty text. Preserve case for display and copy.
resourceUrl	String	No	Optional source URL entered by the user.

Field	Type	Required	Rule
merchantDomain	String	Derived	Canonical registrable domain, for example dropbox.com.
startDate	Date string (ISO)	No	Stored YYYY-MM-DD local calendar date; shown as DD.MM.YYYY.
expiryDate	Date string (ISO)	No	Stored YYYY-MM-DD; shown as DD.MM.YYYY; inclusive through end of local day.
note	String	No	Maximum 140 characters.
notify	Boolean	Yes	Defaults to true only when a usable merchant domain is available; otherwise false. Must be false when no usable merchant domain exists.
createdAt	UTC timestamp	Yes	Creation time for deterministic ordering and troubleshooting.
updatedAt	UTC timestamp	Yes	Updated on every successful edit.

8.2 Date representation

Start Date and Expiry Date are calendar dates, not clock instants. In user-facing forms and lists, display and accept dates as DD.MM.YYYY, for example 31.12.2026. Store them internally as ISO date strings such as 2026-12-31 so comparisons and sorting are reliable. Compare them with the user's current local calendar date. This preserves the agreed rule that an expiry is valid through the end of that day in the current timezone, without daylight-saving changes moving the cutoff. Use UTC timestamps only for events such as createdAt and updatedAt.

8.3 Domain normalization examples

User input or visited host	Canonical key	Match
https://www.dropbox.com/plans	dropbox.com	Yes
dropbox.com	dropbox.com	Yes
shop.dropbox.com	dropbox.com	Yes
dropbox.co.uk	dropbox.co.uk	Only with same registrable domain
notdropbox.com	notdropbox.com	No
dropbox.com.example.org	example.org	No

8.4 Example TypeScript shape

```
type PromoCodeRecord = {
  id: string;
  promoCode: string;
  resourceUrl?: string;
  merchantDomain?: string;
  startDate?: string;   // YYYY-MM-DD, local calendar date
  expiryDate?: string;  // YYYY-MM-DD, inclusive
  note?: string;        // 140 characters maximum
  notify: boolean;
  createdAt: string;    // ISO UTC timestamp
  updatedAt: string;    // ISO UTC timestamp
};
```

The merchantDomain field is derived when a valid resource URL or bare domain is provided. A record without a usable merchant domain can still be stored and managed, but it cannot generate a site-visit reminder until the user adds one.

9. Technical design and stack

9.1 Recommended first-release stack

Layer	Technology	Why it fits
Browser extension	Chrome Manifest V3	Current Chrome extension model, event-driven service worker, toolbar action, optional host permissions.
Language	TypeScript	Strong types for extension APIs, record validation, and maintainable UI state.
UI	React with Vite	Familiar component model and fast local development. Keep the UI small and avoid heavy dependencies.
Local records	chrome.storage.local	Persistent extension-specific storage with ample quota for 100 short records [R1].
Site detection	Optional host access plus dynamically registered content script	Inject only on merchant domains explicitly enabled for reminders [R4, R7].
Reminder UI	chrome.action.openPopup	Opens the PromoBase toolbar popup on a match in Chrome 127 and later [R5, R6].
Account backend	None in first release	Avoids authentication, hosting, and data-protection overhead until local utility is validated.
Later account backend	Python FastAPI and PostgreSQL	Uses Andrii's Python strength; relational records and ownership fit sync and sharing.

9.2 Runtime flow

1. The user saves a record with Notify enabled.
2. The extension explains the requested merchant access and asks Chrome for the optional host permission during the user's save action.
3. If granted, the extension registers a minimal content script for that registrable domain. The script reads the page URL only and sends the domain and tab context to the service worker.
4. The service worker loads matching records from chrome.storage.local, applies start-date and repeat-reminder rules, then opens the action popup.
5. The popup presents the matching record data and copy controls. It does not upload the record or browsing URL to a server.

9.3 Permission baseline

Expected baseline permissions are storage, scripting, and optional host access for supported HTTP and HTTPS merchant sites. The exact manifest must be confirmed during implementation. Avoid broad access to all websites, browsing history, cookies, or tabs unless a demonstrated feature requires it. Chrome recommends the narrowest permissions needed and supports optional permissions requested at runtime [R4].

9.4 Later account phase

When account-backed persistence is justified, add an email-based identity flow, FastAPI service, and PostgreSQL database. The extension remains the client. The server owns users and records, enforces record

ownership, and later supports sync and sharing. Use HTTPS for all requests, short-lived access tokens, secure refresh handling, and account deletion/export flows. Do not store user passwords in extension storage. MongoDB is not recommended for this data model unless requirements materially change.

9.5 Open-source project

PromoBase is intended to be developed in a public GitHub repository. This does not change the local-first data model: a public source repository does not publish users' locally stored promo codes. Before making the repository public, remove credentials, private test data, and personal user records; use synthetic examples only. MIT is the selected open-source license. Add the MIT license file to the repository and identify it in the README before public release. GitHub notes that a public repository without a license remains subject to default copyright restrictions, even though other GitHub users may view and fork it [R14]. Keep dependency licenses and Chrome Web Store privacy disclosures accurate. If the shipped extension handles user data, publish an up-to-date privacy policy describing its collection, use, and sharing [R3, R15].

10. Privacy, security, and permission model

- Local-first: do not send promo codes, notes, domains, or browsing activity to a PromoBase server in the first release.
- Least privilege: ask for site access only for domains with Notify enabled; explain that access is used to show saved codes on visits.
- Minimal page access: the content script needs the current hostname and must not inspect page text, forms, cart contents, passwords, or checkout information.
- No analytics by default: avoid collection of browser history and user behavior in the initial version. Product learning can use voluntary interviews and manual feedback.
- Data retention: local records persist through ordinary browsing-data clearing, but extension removal or Chrome profile loss may delete them [R1, R2].
- Store listing: before Chrome Web Store publication, provide a privacy policy and accurate data-use and permission disclosures if any data handling changes or the store requires disclosures for the shipped feature set [R3].
- Open or restricted pages: Chrome internal pages and some protected pages cannot host extension scripts. The UI should explain that reminders only work on supported web pages.

11. Market comparison and fit hypothesis

PromoBase participates in the broad coupon-extension market but solves a narrower memory problem. The products below are compared by their publicly described core behavior, not by a full feature audit.

Product	Core behavior	Overlap with PromoBase	PromoBase distinction
PayPal Honey	Searches for and tests available coupon codes at checkout; may apply a working code [R8].	Displays code opportunities while the user shops.	PromoBase stores codes the user already encountered, including creator-specific codes, and remembers them across time.
Capital One Shopping	Tests community-sourced codes, compares prices, and offers shopping rewards [R9].	Shows relevant promo opportunities on merchant sites.	PromoBase is a personal saved-code library, not an automatic discovery, price, or rewards service.
Rakuten	Alerts users to cashback, coupons, and deals and supports coupon application at supported stores [R10].	Uses browser extension prompts during shopping.	PromoBase has no affiliate attribution or cashback activation. It surfaces the user's own saved records.
PromoBase first release	Manual save, local organization, and domain-based reminder.	Shares the general goal of helping users remember discounts.	Creator-to-code memory, user-owned records, local-first behavior, and no account requirement.

11.1 Fit score

Current market-fit hypothesis: 4 out of 10. This is a qualitative product judgment, not a measured demand statistic. Coupon use is established, but PromoBase asks users to manually save codes and competes for attention with mature extensions that automate coupon discovery. Its clearest potential segment is people who frequently hear creator-sponsored codes and later shop on the advertised sites.

The concept is worth prototyping because the first version is inexpensive to build and directly tests a specific user habit. A higher score should require evidence that users save codes repeatedly, return to the reminders, and prefer this workflow over notes, bookmarks, or existing coupon extensions.

11.2 Positioning statement

For people who encounter promo codes before they are ready to shop, PromoBase is a personal code library that remembers the merchant and brings the code back when they visit that site. Unlike automatic coupon engines, PromoBase focuses on the codes the user chose to save.

11.3 Build viability

Estimated build viability for a solo developer: 8 out of 10 for the local first release. The core involves standard extension UI, local record management, date rules, domain normalization, and a narrow permission-aware reminder flow. The main learning curve is Chrome extension lifecycle and permission behavior rather than record-management logic. The full account, sync, sharing, and payment vision is a larger product and operations commitment, estimated at 5 out of 10 for solo delivery without reducing scope or using managed services.

12. Success measures and validation plan

The first release should prioritize observed usefulness and reliability over growth or revenue. Because the MVP is local-first and has no analytics, measure through opt-in user interviews, task observations, and local development checks rather than hidden tracking.

Measure	Initial signal	How to assess
Save-flow completion	A new user can save a valid code without assistance.	Observe 5 to 10 first-time users and note errors or abandonment.
Reminder usefulness	Users recognize a matching merchant prompt and can copy the code.	Ask users to save a test code, revisit the domain, and describe whether timing felt helpful.
Permission acceptance	Users understand why one merchant site needs access.	Record consent/decline in a voluntary test session, not by uploading telemetry.
Repeat use	Users save multiple records over several weeks.	Follow up with volunteers and ask how often they used or ignored the extension.
Reliability	No record loss, false domain match, or repeated-popup defect in tested flows.	QA test matrix across supported Chrome versions and date/timezone scenarios.

12.1 Early validation questions

- How often do people encounter codes before they intend to buy?
- Would they save a code manually, or is that extra step too much friction?
- Is a toolbar popup helpful or disruptive when arriving at a merchant site?
- Will users grant site access when the purpose is limited and clearly explained?
- Do the creator name, source video URL, or product category need to be captured in the first version?

13. Risks and mitigations

Risk	Impact	Mitigation
Automatic popup feels intrusive.	Users disable or uninstall PromoBase.	One popup per merchant visit, easy dismissal, and observe behavior in prototype sessions.
Site permission prompts cause drop-off.	Reminders are not enabled for saved records.	Request access only when Notify is enabled; explain the reason and support later opt-in.
Domain parsing creates false matches.	Wrong codes appear on unrelated sites or expected reminders fail.	Use Public Suffix List-aware parsing and acceptance cases for subdomains and deceptive suffixes.
Guest records are lost after extension removal or profile loss.	User loses saved codes.	Persistent warning; document cloud account and export/import as future recovery options.
Codes are expired or restricted by merchant terms.	User trusts a code that fails at checkout.	Keep expiry warnings visible and state that PromoBase does not verify code validity.
Single developer maintenance burden.	Security and Chrome API changes can break reminders.	Keep permissions narrow, dependencies limited, and test the supported Chrome range before releases.

14. Delivery approach

A staged build reduces unknowns and keeps the first deliverable testable before implementing the full extension workflow.

Stage	Deliverable	Exit condition
1. Product skeleton	Manifest V3 extension, popup shell, local record schema.	Loads unpacked in Chrome and can display the popup.
2. Local record management	Create, list, edit, delete, validation, capacity handling.	All record operations pass acceptance criteria using local storage.
3. Domain matching	Normalization utility and domain test cases.	Root, www, subdomain, public suffix, and deceptive-domain cases resolve correctly.
4. Reminder proof	Per-domain permission, dynamic script, action popup and deduplication.	One correct popup appears on an allowed matching visit and not on a non-matching site.
5. Polish and publish prep	Accessibility, empty/error states, privacy copy, Web Store assets.	Manual QA and listing disclosures are ready for review.

15. Future improvements

The following ideas are intentionally optional and low priority. They should be revisited after the local save-and-remind workflow has been tried with real users.

15.1 Account and durable storage

Allow email-based account creation and synchronize records to a backend so records can survive extension removal, profile reset, or use on another device. Include export, account deletion, and recovery behavior. Candidate stack: Python FastAPI, PostgreSQL, and a managed email identity provider.

15.2 Sharing

Allow a user to share selected codes with another PromoBase user by email address or PromoBase user ID. Define recipient confirmation, revocation, duplicate behavior, sender/recipient ownership, and audit history before implementation.

15.3 Monetization and support

Consider donations and a paid plan after ongoing value is established. Potential subscription benefit: an ad-free experience. Payment processor, cancellation flow, refunds, taxes, and store policy requirements remain undecided. Avoid embedded ads in the first release.

15.4 Product enhancements

- Source context such as creator name, video URL, platform, and date discovered.
- Optional tags, categories, or redeemed status.
- Export and import for local users before cloud accounts exist.
- Reminder style settings, including toolbar popup, in-page card, and standard system notification where supported.
- Support for other browsers after the Chrome experience is stable.

15.5 User feedback

Consider a small Provide Feedback action with an optional five-star rating and free-form comment to support early product learning. Defer direct email submission until there is a secure and privacy-transparent delivery method. A public extension package must not contain email-service credentials. A future implementation should state what information will be sent, ask for consent before submission, send only the rating and comment (plus any explicitly optional diagnostic details), protect the endpoint against spam, and identify the receiving support address or service in the privacy notice. A hosted feedback form can be evaluated as a lower-maintenance alternative to operating a custom email backend.

16. Decisions and assumptions

Topic	Initial decision
Account	No account required for first release.
Storage warning	Records are local to the Chrome profile; extension removal or profile loss may delete them.
Record cap	100 records for guest users.
Record entry	Manual entry only.
Use case	General coupon organizer with emphasis on creator codes personally encountered.
Domain matching	Use registrable domain; root and all subdomains match.
Reminder surface	Open PromoBase's Chrome toolbar popup, not a system notification, for the initial prototype.
Dates	Date-only values. Expiry is inclusive to the end of the date in the current local timezone.
Expired records	Retained, grayed out, sorted below active records, and still shown with a warning on visit.
Undated records	Remain in the extension indefinitely.
Stack	TypeScript, Manifest V3, React/Vite, chrome.storage.local. Python/FastAPI/PostgreSQL later.
Market score	4/10 qualitative hypothesis pending user validation.

Topic	Initial decision
Open source	Public GitHub repository under MIT. Add the license file before public release.
Date format	Display and accept DD.MM.YYYY. Persist ISO YYYY-MM-DD internally.
Notify default	Checked only when a usable Resource or Product URL exists; otherwise unchecked and disabled.
Save failures	Preserve input, explain the issue, and never claim a failed write succeeded.
Feedback form	Potential low-priority follow-up. Requires consent, privacy disclosure, and secure delivery.

16.1 Assumptions to keep visible

- Start Date means the code should not generate a reminder before that local calendar date.
- A record with Resource or Product URL left empty can still be saved, but cannot trigger a site-visit reminder.
- If several codes match one merchant, PromoBase displays them together in a single popup.
- Codes are user-provided and are not verified by PromoBase. Merchant restrictions may apply.
- Chrome 127 or later is the minimum proposed version for programmatic action popup support.

17. Glossary

Term	Meaning
Promo code	A text code a user can enter at a merchant to request a discount or promotion.
Merchant domain	The registrable domain used to associate a record with a merchant website, such as dropbox.com.
Registrable domain	The domain a person can register under a public suffix, derived with Public Suffix List rules.
Host or hostname	The host portion of a URL, such as www.dropbox.com or shop.dropbox.com.
Manifest V3	The current Chrome extension architecture and manifest format used for the proposed build.
Optional host permission	A site access permission that an extension can request later for a specific host rather than requesting access to all sites at install time.
Guest mode	Using PromoBase without an account, with records stored in the current browser profile.

18. Additional Notes

19. References

Platform behavior and competitor descriptions were checked against the following official documentation and product help pages on 8 October 2026. Product feature descriptions may change over time.

R1. Chrome for Developers. chrome.storage API. Open source
R2. Chrome for Developers. Storage and cookies in extensions. Open source
R3. Chrome Web Store. User data privacy and secure handling requirements. Open source
R4. Chrome for Developers. Permissions API and optional permissions. Open source
R5. Chrome for Developers. Action API. Open source
R6. Chrome for Developers. Add a popup. Open source
R7. Chrome for Developers. Content scripts. Open source
R8. PayPal Honey Help. What is the Honey extension. Open source
R9. Capital One Shopping Help Center. Open source
R10. Rakuten Help. Browser Extension Help. Open source
R11. Chrome for Developers. Manifest file format. Open source
R12. Chrome Web Store. Program policies. Open source
R13. Public Suffix List. Open source
R14. GitHub Docs. Licensing a repository. Open source
R15. Chrome Web Store. Program policies. Open source